

MINISTRY OF EDUCATION AND TRAINING
UNIVERSITY OF ECONOMICS AND FINANCE
HO CHI MINH CITY



PROJECT

ANNUAL INITIATIVE EVALUATION SYSTEM

Performed by Student:

Nguyễn Ngọc Triệu	22D1ITqE-SWE02	CNTT
Nguyễn Hoàng Phúc	22D1ITE-SWE02	CNTT
Phạm Quang Vũ	22D1ITE-SWE02	CNTT
Nguyễn Tấn Thiện Huy	22D1ITE-SWE02	CNTT

Ho Chi Minh City, April 2026.

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Ho Chi Minh City, April 2026.

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CHAPTER 1. OVERVIEW

1.1. Project information

IdeaTrack is a web-based Initiative Management System developed using ASP.NET Core MVC (.NET 8), PostgreSQL, and Entity Framework Core. The system is designed to digitalize and standardize the workflow of academic initiatives in a university environment, where proposals must pass through several controlled review stages before final approval. By centralizing initiative submission, review, scoring, and decision-making, IdeaTrack helps reduce manual handling, improve traceability, and enhance the consistency of the approval process.

From a functional perspective, the system supports a complex state transition workflow that reflects real-world academic governance. An initiative may move through multiple statuses such as draft creation, submission, faculty-level review, scientific committee evaluation, final processing, and final approval or rejection. Because each transition depends on both the current state and the user role, the system requires strict authorization control and logical validation to ensure that only qualified users can perform permitted actions at the correct stage.

Within the scope of this testing project, five main modules are included: Authentication, Author, Faculty Leader, Councils, and SciTech Admin. These modules together form the core business flow of the application. In addition, IdeaTrack is not only a conventional website but also integrates internal API mechanisms to support secure and efficient state management across its layered workflow. Therefore, the testing focus is placed on validating role-based access control, UI behavior, business rules, and the correctness of state transitions throughout the initiative lifecycle.

1.2. Project team

No.	Full Name	Role & Task Allocation
1	Nguyễn Ngọc Triệu	Team Leader – responsible for Test Plan coordination, report integration, progress tracking, and final quality review. Also executes 10 manual test cases and participates in testing the Authentication module.
2	Phạm Quang Vũ	QA Automation & Module Tester – responsible for designing and executing 10 manual test cases, developing Selenium-based automation scripts for bonus points, and

		testing the Author module.
3	Nguyễn Hoàng Phúc	Bug Manager & Module Tester – responsible for managing JIRA bug tracking, classifying Severity/Priority, executing 10 manual test cases, and testing the Faculty Leader module.
4	Nguyễn Tấn Thiện Huy	System Architect & Module Tester – responsible for analyzing state transition logic, preparing workflow diagrams, executing 10 manual test cases, and testing the Councils and SciTech Admin modules.

Table Error: Reference source not found-1 Team members and roles

All four team members contribute equally to manual testing, defect reporting, and documentation. Each member is assigned one primary functional area while still sharing responsibility for overall test design, execution, and report writing to ensure balanced workload distribution and comprehensive coverage of the system scope.

CHAPTER 2. TEST PLAN

2.1. Test scope

2.1.1. *In-scope*

The system testing phase is structured around specialized functional modules to ensure comprehensive validation of all business processes. The primary focus of the testing scope encompasses the following core areas:

System Administration and Security Testing begins with a rigorous verification of the Authentication Module and Role-Based Access Control (RBAC). This ensures secure user login and logout processes while validating that access and state transitions are strictly governed by assigned roles and business rules. Concurrently, the SciTech Admin Module is evaluated to confirm the system's foundational stability, focusing on period and academic year management, template configuration, user administration, and the proper handling of final-stage initiative processing.

Initiative Processing and Evaluation Workflows The core user operations are systematically validated starting with the Author Module, which covers initiative creation, draft saving, file attachments, co-author management, and tracking functions. The review and evaluation stages are then tested through the Faculty Leader Module, ensuring correct role-restricted decision-making for approvals and rejections at the faculty level. Following this, the Councils Module undergoes testing for council assignments, score submissions, re-evaluation requests, and final score consolidation to guarantee a transparent and accurate assessment workflow.

State Transition and Lifecycle Management A critical area of focus during testing is the State Transition Workflow of Initiatives. The testing team traces the complete initiative lifecycle, tracking its progression from *Draft* through intermediate states (*Pending*, *Faculty_Approved*, *Evaluating*, *Pending_Final*, *Processing*) to the final outcomes (*Approved* or *Rejected*). To ensure system resilience and flexibility, the testing scope also comprehensively covers rollback and recovery mechanisms, thoroughly validating the *Clone*, *Revoke*, and *Resubmit* operational flows.

2.1.2. *Out-of-scope*

Due to time constraints and the academic scope of the project, the following activities are excluded from testing:

- Performance testing / Stress testing

- Advanced security penetration testing
- Mobile responsiveness testing on mobile devices

2.2. Test strategy & approach

The test strategy for IdeaTrack is designed according to a black-box testing-oriented approach, supported by selective white-box review and limited automation for high-value scenarios. Since the system is primarily a workflow-driven academic platform, the main testing objective is to verify that business rules, role permissions, and state transitions operate correctly under both normal and negative conditions.

2.2.1. *Black-box testing*

The team applies Equivalence Partitioning (EP) and Boundary Value Analysis (BVA) to form-based inputs in critical modules. For example, the initiative creation form is tested with valid and invalid input groups such as empty fields, maximum-length values, and malformed data. Similarly, the council scoring form, which uses a 100-point scale, is tested at boundary conditions such as minimum score, maximum score, and values around critical limits. This approach helps detect input validation defects efficiently while minimizing redundant test cases.

In addition, the project applies State Transition Testing as a core design technique because IdeaTrack is fundamentally a workflow-based system. Each initiative moves through a controlled sequence of statuses, and every transition depends on both the current state and the active user role. Therefore, the team verifies not only the normal progression of states, such as Pending → Evaluating → Processing, but also alternate and exceptional flows, including rollback, re-evaluation, resubmission, and revocation. This method is essential for detecting logical defects in the transition rules, such as unauthorized state changes, skipped stages, incorrect rejection handling, or inconsistent recovery behavior after a decision is reversed.

2.2.2. *White-box testing*

Although the project is primarily black-box in nature, the team also performs limited white-box review to improve understanding of system implementation before execution. Specifically, the team reviews key Controller files and the DataSeederService.cs source code in the ASP.NET Core project to understand how data is initialized, how entities are linked, and how the application enforces workflow rules at the code level. This review

helps the testers interpret UI behavior correctly, verify seeded test data, and prepare more accurate test conditions for each module.

2.2.3. *Manual and automation*

The testing process combines manual and automated testing.

Manual Testing: The team executes manual test cases to cover the principal business flows across the five core modules. Manual testing is used for role-specific actions, workflow transitions, validation errors, UI behavior, and negative test scenarios. All defects discovered during manual execution are logged in JIRA with strict Severity and Priority classification to support defect triage and follow-up tracking.

Automation Testing: To enhance repeatability, the team uses Selenium WebDriver and C# (xUnit) to automate selected happy-path scenarios such as login and initiative creation. These automated scripts help validate stable workflows and generate an HTML test report for result documentation.

2.3. Test environment

Item	Specification
Operating System	Windows 10/11, macOS
Browsers	Google Chrome (Latest), Microsoft Edge (Latest)
Application Environment	Kestrel Web Server (.NET 8), localhost:5227
Database	PostgreSQL
Test Management & Bug Tracking	JIRA
Automation Tools	xUnit, Selenium WebDriver

Table Error: Reference source not found-1 Environment details

The above environment was selected to ensure consistency with the development stack of IdeaTrack and to provide stable conditions for both manual execution and automation-based verification.

CHAPTER 3. Test Design & Execution

3.1. Test scenarios

Scenario ID	Module	Scenario Description	Priority
SC-01	Homepage Features	Verification of general landing page elements, public announcements, system guidelines, and basic navigation before login.	Medium
SC-02	Author Module	Validation of initiative creation, draft saving, file attachment upload, co-author management, and submission tracking.	High
SC-03	Faculty Leader Module	Testing of initiative review at the faculty level, including approval, rejection, and role-restricted decision-making.	High
SC-04	SciTech Admin Module	Verification of period/academic year management, template configuration, user administration, and final-stage initiative processing.	Critical
SC-05	Council Member Module	Validation of council assignment viewing, score submission, evaluation workflow, and re-evaluation processing.	High
SC-06	Initiative State Flow	End-to-end testing of the full initiative lifecycle (Draft to Approved/Rejected), including rollback flows (Clone, Revoke, Resubmit).	Critical

Table Error: Reference source not found-1 Test scenarios details

3.2. Test case specification

The team has designed and executed more than 40 detailed test cases covering both UI and API-level behaviors. The following tables present the standard test case structure and several representative samples. The complete set of test cases is attached in Excel format in the Appendix section.

3.2.1. Author Module

The screenshot shows the 'Create Initiative' page in the 'Initiative Portal'. The page has a red header with the portal name and user controls. A sidebar on the left lists navigation options: Home, Leaderboard, About, Recognized Initiatives, Guidelines, and a 'Register Initiative' button. The main content area is titled 'Create Initiative' and contains a form with the following fields:

- Academic Year ***: A dropdown menu showing 'Academic Year 2025-2026'.
- Category ***: A dropdown menu with the placeholder '-- Select academic year first --'.
- Department**: A text input field containing 'Department of Information Technology'.
- Budget (VND)**: A text input field containing '0'.
- Initiative Title ***: A text input field with the placeholder 'Enter a brief title for the initiative'.
- Co-Authors (Optional)**: A field for adding co-authors.

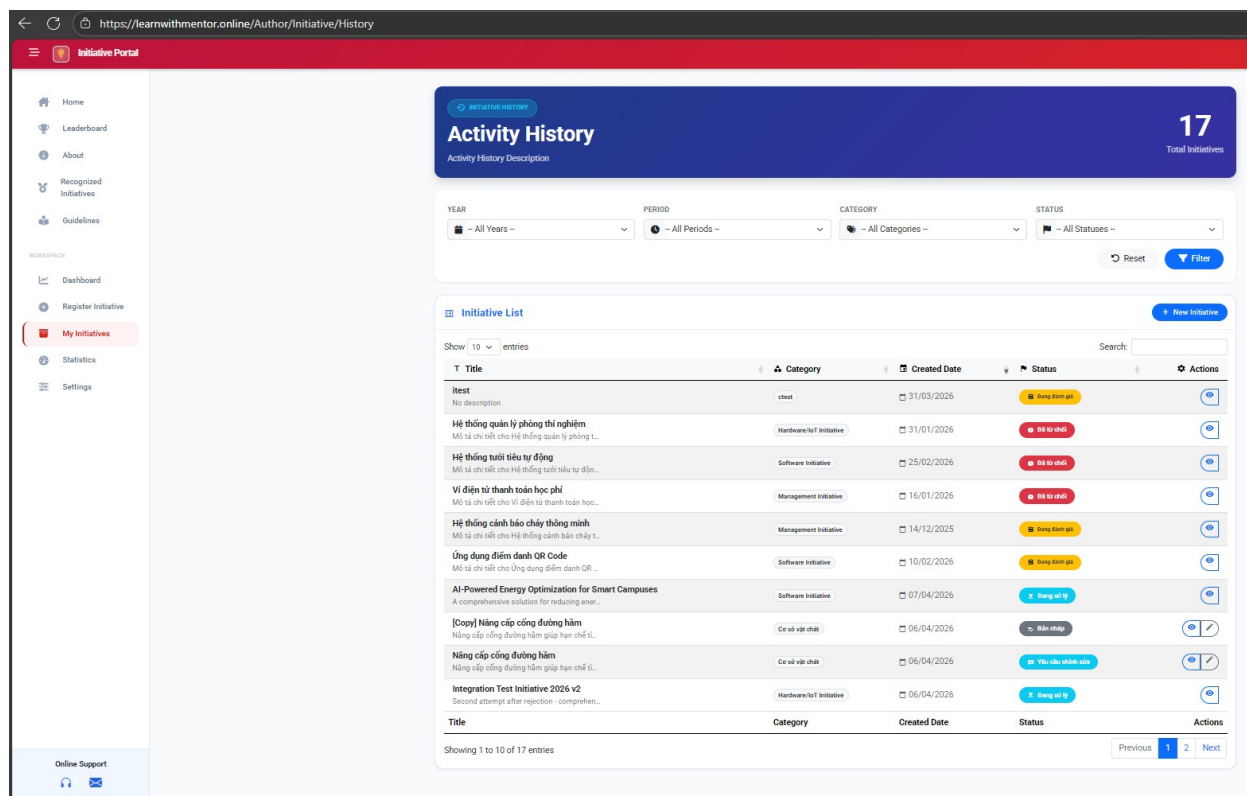
Picture Error: Reference source not found-1 Function author create initiative

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
AU-CI-01	Create Initiative — Valid Data (EP: valid partition)	User is logged in as Author. Academic Year and Category exist.	1. Navigate to Register Initiative. 2. Fill: Title = "AI Solution", AcademicYearId = 1, CategoryId = 2, File = "tailieu.pdf", CoAuthorIds = [2, 3]. 3. Click Submit.	System saves the initiative successfully and redirects to My Initiatives page.	PASS
AU-CI-02	Create Initiative — Blank Title (EP: invalid — empty string)	User is logged in as Author.	1. Navigate to Register Initiative. 2. Leave Title blank, fill other fields. 3. Click Submit.	Validation error: "The Title field is required." Initiative is not saved.	PASS
AU-CI-03	Create Initiative — Missing Academic Year (EP: invalid — null selection)	User is logged in as Author.	1. Navigate to Register Initiative. 2. Fill Title, do NOT select Academic Year. 3. Click Submit.	Submit button is disabled or server rejects with validation error "Please select an Academic Year."	PASS
AU-CI-04	Create Initiative — Invalid File Type (EP: invalid file extension)	User is logged in as Author.	1. Navigate to Register Initiative. 2. Fill valid data, attach File = "script.exe". 3. Click Submit.	Validation error: "Invalid file format, only (.pdf, .docx, ...) allowed." File is rejected.	PASS

AU-CI-05	Create Initiative — File Size Exceeded (BVA: above max limit)	User is logged in as Author. Max file size = 10 MB.	1. Navigate to Register Initiative. 2. Fill valid data, attach File = "video.mp4" (50 MB). 3. Click Submit.	Validation error: "File size exceeds the allowed limit (10 MB)." Submission blocked.	PASS
AU-CI-06	Create Initiative — File Size at Boundary (BVA: exactly 10 MB)	User is logged in as Author. Max file size = 10 MB.	1. Navigate to Register Initiative. 2. Fill valid data, attach a file exactly 10 MB. 3. Click Submit.	File is accepted and initiative is created successfully.	PASS
AU-CI-07	Create Initiative — File Size Just Below Limit (BVA: 9.99 MB)	User is logged in as Author. Max file size = 10 MB.	1. Navigate to Register Initiative. 2. Fill valid data, attach a file of 9.99 MB. 3. Click Submit.	File is accepted and initiative is created successfully.	PASS
AU-CI-08	Create Initiative — File Size Just Above Limit (BVA: 10.01 MB)	User is logged in as Author. Max file size = 10 MB.	1. Navigate to Register Initiative. 2. Fill valid data, attach a file of 10.01 MB. 3. Click Submit.	Validation error: "File size exceeds the allowed limit (10 MB)."	PASS
AU-CI-09	Create Initiative — Title at Min Length (BVA: 1 character)	User is logged in as Author.	1. Navigate to Register Initiative. 2. Enter Title = "A". 3. Fill other required fields. 4. Click Submit.	Initiative is created successfully with single-character title.	PASS
AU-CI-10	Create Initiative — Title at Max Length (BVA: max allowed chars)	User is logged in as Author.	1. Navigate to Register Initiative. 2. Enter Title with maximum allowed characters (e.g., 255 chars). 3. Fill other fields. 4. Click Submit.	Initiative is created successfully. Title is stored and displayed correctly.	PASS
AU-CI-11	Create Initiative — Title Exceeds Max Length (BVA: max + 1)	User is logged in as Author.	1. Navigate to Register Initiative. 2. Enter Title with max + 1 characters. 3. Click Submit.	Validation error or title is truncated. Initiative is not saved with oversized title.	PASS
AU-	Create Initiative	User is logged in	1. Navigate to Register	Initiative is created	PASS

CI-12	— No File Attached (EP: optional file = empty)	as Author. File is optional.	Initiative. 2. Fill Title, Academic Year, Category. Do NOT attach file. 3. Click Submit.	successfully without attachment.	
AU-CI-13	Create Initiative — Missing Category (EP: invalid — null)	User is logged in as Author.	1. Navigate to Register Initiative. 2. Fill Title, select Academic Year. Do NOT select Category. 3. Click Submit.	Validation error: "Category is required."	PASS
AU-CI-14	Create Initiative — Budget = 0 (BVA: lower bound)	User is logged in as Author.	1. Navigate to Register Initiative. 2. Fill all required fields, set Budget = 0. 3. Click Submit.	Initiative is created with Budget = 0.	PASS
AU-CI-15	Create Initiative — Negative Budget (EP: invalid numeric)	User is logged in as Author.	1. Navigate to Register Initiative. 2. Fill all fields, set Budget = -1000. 3. Click Submit.	Validation error: "Budget must be a non-negative number."	PASS

Table Error: Reference source not found-2 Create initiative testcases

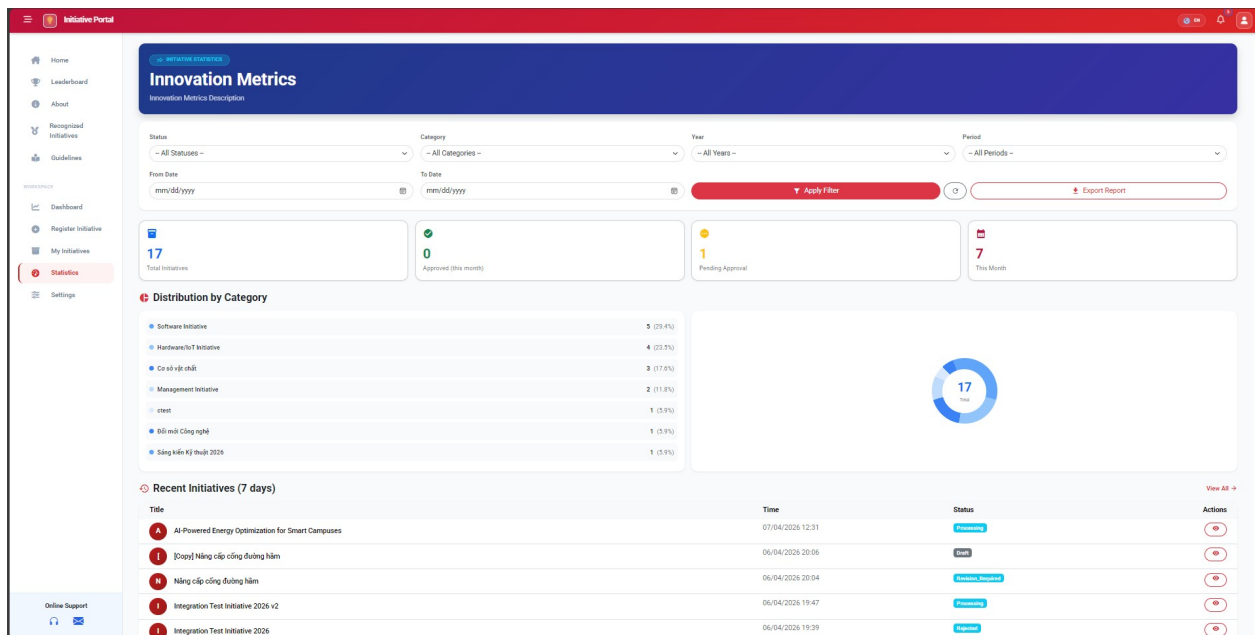


Picture Error: Reference source not found-2 Function author view activity history

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
AU-MI-01	View My Initiatives List (EP: valid view)	Author is logged in. Author has submitted initiatives.	1. Navigate to My Initiatives.	Initiative list is displayed showing all author's initiatives with Title, Category, Created Date, Status, and Actions columns. Showing 1-10 of 17 entries.	PASS
AU-MI-02	Filter Initiatives by Year (EP: valid filter)	Author is logged in. Multiple academic years exist.	1. Navigate to My Initiatives. 2. Select Year dropdown. 3. Click "Filter".	Only initiatives from the selected academic year are displayed.	PASS
AU-MI-03	Filter Initiatives by Period (EP: valid filter)	Author is logged in. Multiple periods exist.	1. Navigate to My Initiatives. 2. Select Period dropdown. 3. Click "Filter".	Only initiatives from the selected period are displayed.	PASS
AU-MI-04	Filter Initiatives by Category	Author is logged in.	1. Navigate to My Initiatives.	Only initiatives with the selected category	PASS

	(EP: valid filter)	Multiple categories exist.	2. Select Category dropdown (e.g., "Software Initiative"). 3. Click "Filter".	are displayed.	
AU-MI-05	Filter Initiatives by Status (EP: valid filter)	Author is logged in. Initiatives with various statuses exist.	1. Navigate to My Initiatives. 2. Select Status dropdown (e.g., "Rejected"). 3. Click "Filter".	Only initiatives with "Rejected" status are displayed.	PASS
AU-MI-06	Search Initiatives in List (EP: valid search)	Author is logged in. Initiatives exist.	1. Navigate to My Initiatives. 2. Type "AI" in the Search box.	Only initiatives with "AI" in their title are shown.	PASS
AU-MI-07	Search with Non-Existent Keyword (EP: no match)	Author is logged in.	1. Navigate to My Initiatives. 2. Type "XYZNOTEXIST" in Search box.	No results displayed. Table shows empty or "No data available" message.	PASS
AU-MI-08	Reset All Filters (EP: reset action)	Author is logged in. Filters are currently applied.	1. Navigate to My Initiatives. 2. Apply some filters. 3. Click "Reset" button.	All filters are cleared. Full initiative list is displayed.	PASS
AU-MI-09	Pagination — Navigate to Page 2 (BVA: page boundary)	Author is logged in. More than 10 initiatives exist (17 total).	1. Navigate to My Initiatives. 2. Click page "2" or "Next" in pagination.	Page 2 displays entries 11-17. Pagination updates correctly.	PASS
AU-MI-10	Change Entries Per Page (BVA: display count)	Author is logged in. 17 initiatives exist.	1. Navigate to My Initiatives. 2. Change "Show entries" dropdown from 10 to 25.	All 17 initiatives are displayed on single page.	PASS
AU-MI-11	Edit Draft Initiative (EP: valid edit)	Author is logged in. Draft initiative exists.	1. Navigate to My Initiatives. 2. Click edit icon on a "Draft" initiative.	Initiative edit form is opened. Author can modify fields and re-submit.	PASS
AU-MI-12	View Initiative Details (EP: valid view)	Author is logged in. Initiative exists.	1. Navigate to My Initiatives. 2. Click view icon on any initiative.	Initiative detail page is displayed with full information.	PASS

Table Error: Reference source not found-3 Activity history testcases

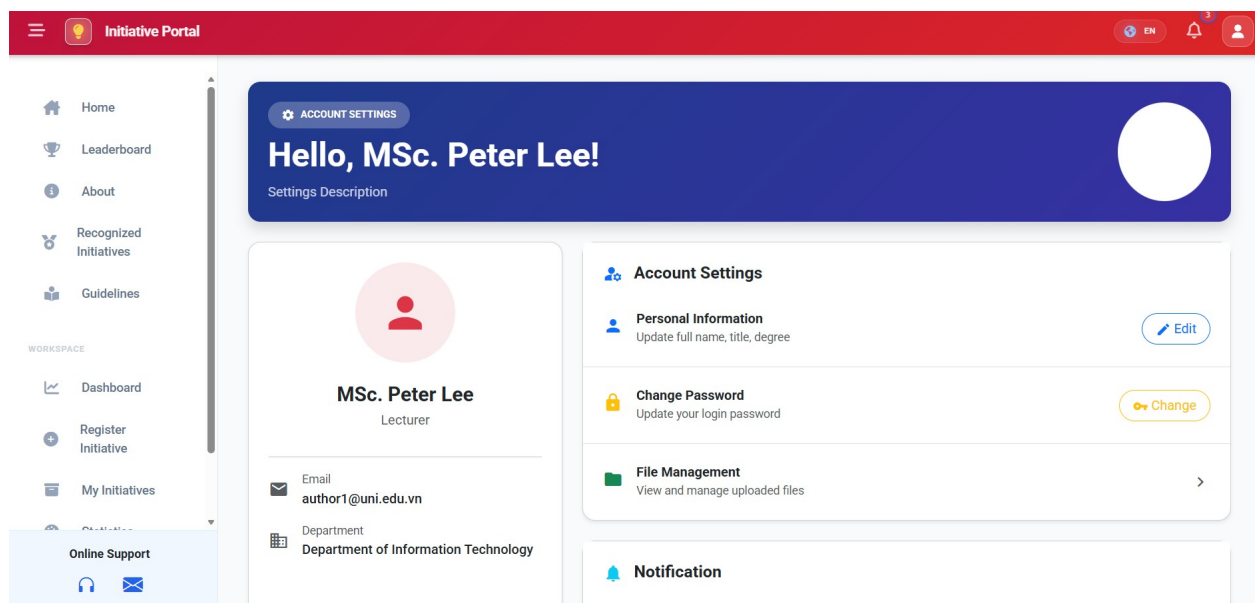


Picture Error: Reference source not found-3 Function author statistic

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
AU-ST-01	View Statistics Dashboard (EP: valid view)	Author is logged in. Author has submitted initiatives.	1. Navigate to Statistics page.	Statistics page displays: Total Initiatives = 17, Approved (this month) = 0, Pending Approval = 1, This Month = 7.	PASS
AU-ST-02	View Distribution by Category Chart (EP: valid chart)	Author is logged in. Multiple categories used.	1. Navigate to Statistics page. 2. View "Distribution by Category" section.	Pie chart and list show correct distribution: Software Initiative 5 (29.4%), Hardware/IoT 4 (23.5%), etc.	PASS
AU-ST-03	View Recent Initiatives List (EP: valid list)	Author is logged in. Recent initiatives exist.	1. Navigate to Statistics page. 2. View "Recent Initiatives (7 days)" section.	Recent initiatives are listed with Title, Time, Status, and Actions. Correct statuses shown (Processing, Draft, Revision_Required, etc.).	PASS
AU-ST-04	Filter Statistics by Status (EP: valid filter)	Author is logged in.	1. Navigate to Statistics. 2. Select Status filter (e.g., "Processing"). 3. Click "Apply Filter".	Statistics and charts update to show only data for "Processing" status.	PASS
AU-ST-05	Filter Statistics by Category (EP: valid filter)	Author is logged in.	1. Navigate to Statistics. 2. Select Category filter (e.g., "Software	Statistics update to show only Software Initiative data.	PASS

			Initiative"). 3. Click "Apply Filter".		
AU-ST-06	Filter Statistics by Year (EP: valid filter)	Author is logged in.	1. Navigate to Statistics. 2. Select Year filter. 3. Click "Apply Filter".	Statistics update to show only data from the selected year.	PASS
AU-ST-07	Filter Statistics by Date Range (BVA: valid range)	Author is logged in.	1. Navigate to Statistics. 2. Set From Date and To Date. 3. Click "Apply Filter".	Statistics update to show only data within the date range.	PASS
AU-ST-08	Filter with From Date > To Date (BVA: invalid range)	Author is logged in.	1. Set From Date = "2026-05-01". 2. Set To Date = "2026-04-01". 3. Click "Apply Filter".	Validation error or no results returned. System handles gracefully.	PASS
AU-ST-09	Export Statistics Report (EP: valid export)	Author is logged in. Statistics data exists.	1. Navigate to Statistics. 2. Click "Export Report" button.	Report file is downloaded containing author's initiative statistics.	PASS

Table Error: Reference source not found-4 Author statistic testcases



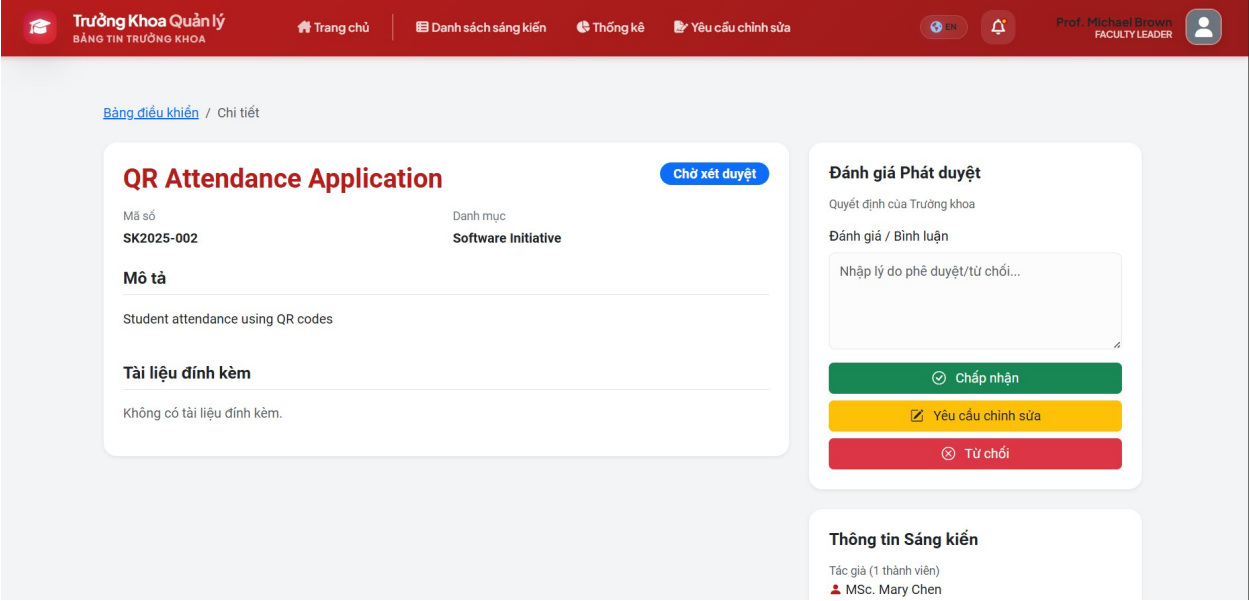
Picture Error: Reference source not found-4 Function author update profile

ID	Title	Pre-conditions	Steps	Expected / Actual	Status
----	-------	----------------	-------	-------------------	--------

				Results	
AU-PF-01	Update Profile — Valid Data (EP: valid partition)	User is logged in as Author (MSc. Peter Lee).	1. Navigate to Settings > Personal Information > Edit. 2. Set FullName = "Nguyen Van A", Position = "Lecturer", Degree = "Master". 3. Click Save.	Profile is updated successfully. Notification: "Update successful."	PASS
AU-PF-02	Update Profile — Blank FullName (EP: invalid — empty)	User is logged in as Author.	1. Navigate to Settings > Personal Information > Edit. 2. Clear FullName field. 3. Click Save.	Validation error: "Full name is required." Database state unchanged.	PASS
AU-PF-03	Update Profile — FullName with Special Characters (EP: edge)	User is logged in as Author.	1. Navigate to Settings > Edit. 2. Set FullName = "Dr. O'Brien-Smith Jr.". 3. Click Save.	Profile is updated. Special characters are stored and displayed correctly.	PASS
AU-PF-04	Update Profile — Very Long FullName (BVA: max length)	User is logged in as Author.	1. Navigate to Settings > Edit. 2. Enter FullName with 255 characters. 3. Click Save.	Profile is updated successfully or shows max-length validation error.	PASS

Table Error: Reference source not found-5 Author profile testcases

3.2.2. Faculty module



Picture Error: Reference source not found-5 Function faculty review initiative

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
FA-RV-01	Review — Accept Initiative (EP: valid accept action)	Faculty user logged in. Initiative status = Submitted. Initiative exists (e.g., SK2025-002).	1. Navigate to Initiative List. 2. Open initiative detail. 3. Enter comment: "Good initiative." 4. Click "Accept".	Status changes to Faculty_Approved. System notification: "Initiative accepted..."	PASS
FA-RV-02	Review — Request Revision with Comment (EP: valid revision request)	Faculty user logged in. Initiative status = Submitted.	1. Open initiative detail. 2. Enter comment: "Please add more data." 3. Click "Request Revision".	Status changes to Revision_Required. System records revision request and notifies author.	PASS
FA-RV-03	Review — Request Revision with Blank Comment (BVA: empty comment boundary)	Faculty user logged in. Initiative status = Submitted.	1. Open initiative detail. 2. Leave comment field blank. 3. Click "Request Revision".	If comment is required: validation error. If optional: revision request is submitted.	PASS
FA-RV-04	Review — Reject Initiative (EP: valid reject)	Faculty user logged in. Initiative status =	1. Open initiative detail. 2. Enter comment:	Status changes to Rejected. System notification: "Initiative has been rejected."	PASS

	action)	Submitted.	"Not feasible." 3. Click "Reject".		
FA-RV-05	Review — Accept Without Comment (EP: optional comment)	Faculty user logged in. Initiative status = Submitted.	1. Open initiative detail. 2. Leave comment empty. 3. Click "Accept".	Status changes to Faculty_Approved. Comment is stored as empty.	PASS
FA-RV-06	Review — Reject Without Comment (BVA: empty comment)	Faculty user logged in. Initiative status = Submitted.	1. Open initiative detail. 2. Leave comment empty. 3. Click "Reject".	Validation error if comment is required for rejection, else status changes to Rejected.	PASS
FA-RV-07	Review — Comment with Max Length (BVA: upper bound)	Faculty user logged in. Initiative status = Submitted.	1. Open initiative detail. 2. Enter comment with maximum allowed characters. 3. Click "Accept".	Comment is accepted and stored. Status changes to Faculty_Approved.	PASS

Table Error: Reference source not found-6 Faculty review testcases

Trường Khoa Quản lý

BẢNG TIN TRƯỞNG KHOA

Trang chủ

Danh sách sáng kiến

Thống kê

Yêu cầu chỉnh sửa

EN

🔔

Prof. Michael Brown

FACULTY LEADER

Bảng điều khiển Trường Khoa

Tổng quan về các sáng kiến của khoa và đánh giá phê duyệt

Hôm nay: 11 April 2026

CHỖ PHÊ DUYẾT

2

KHOA ĐÃ DUYẾT

0

ĐANG ĐÁNH GIÁ

2

CHỖ CHUNG KHẢO

0

ĐÃ DUYẾT

0

BỊ TỪ CHỐI

3

Year

Period

Category

Status

Search by title, code...

Filter

Reset

SÁNG KIẾN	TÁC GIẢ	DANH MỤC	TRẠNG THÁI	NGÀY NỘP	THAO TÁC
AI-Powered Energy Optimization for Smart Campuses SK-2026-0063	MSc. Peter Lee (1 thành viên)	Software Initiative	Processing	07/04/2026	Chi tiết
Smart Library Management System SK2025-001	MSc. Peter Lee (2 thành viên)	Cơ sở vật chất	Processing	06/04/2026	Chi tiết
Nâng cấp cổng đường hầm SK-2026-0061	MSc. Peter Lee (2 thành viên)	Cơ sở vật chất	Yêu cầu chỉnh sửa	06/04/2026	Chi tiết
Integration Test Initiative 2026 v2 SK-2026-0060	MSc. Peter Lee (1 thành viên)	Hardware/IoT Initiative	Processing	06/04/2026	Chi tiết
Integration Test Initiative 2026	MSc. Peter Lee				

Picture Error: Reference source not found-6 Function faculty dashboard

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
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FA-DS-01	Search Initiatives by Keyword (EP: valid keyword)	Faculty user logged in. Initiatives exist.	1. Navigate to Dashboard. 2. Enter search string "AI Solution" in search bar. 3. Click Search / Apply.	List displays only initiatives matching "AI Solution."	PASS
FA-DS-02	Filter by Academic Year / Period (EP: valid filter)	Faculty user logged in. Multiple years/periods exist.	1. Navigate to Dashboard. 2. Select Year = "2025-2026", Period = "2025-2026". 3. Click Apply / Filter.	Initiative list and statistics charts update to show only data for the selected year/period.	PASS
FA-DS-03	Export Dashboard to Excel (EP: valid export)	Faculty user logged in. Initiative data exists.	1. Navigate to Dashboard. 2. Click "Export" button.	Browser downloads CSV file "Initiative_Stats_ddMMyyyy.csv" with correct faculty data.	PASS
FA-DS-04	View File as PDF (EP: valid file conversion)	Faculty user logged in. Initiative has attachment "tailieu.docx".	1. Navigate to initiative detail. 2. Click "View File" on attached document.	System converts docx to PDF via LibreOffice and opens in browser PDF viewer.	PASS
FA-DS-05	Search with Empty Keyword (BVA: empty string)	Faculty user logged in.	1. Navigate to Dashboard. 2. Leave search bar empty. 3. Click Search.	All initiatives are displayed (no filter applied).	PASS
FA-DS-06	Search with Non-Existent Keyword (EP: no match)	Faculty user logged in.	1. Navigate to Dashboard. 2. Enter "XYZNOTEXIST999". 3. Click Search.	No results found. Appropriate "No results" message displayed.	PASS
FA-DS-07	Search with Special Characters (EP: edge — SQL injection)	Faculty user logged in.	1. Enter search string: "'; DROP TABLE --". 2. Click Search.	No results or safe handling. No database error or crash.	PASS

Table Error: Reference source not found-7 Faculty dashboard testcases

Trường Khoa Quản lý
BẢNG TIN TRƯỞNG KHOA

[Trang chủ](#)
[Danh sách sáng kiến](#)
[Thống kê](#)
[Yêu cầu chỉnh sửa](#)

EN

Prof. Michael Brown
FACULTY LEADER

Chỉnh sửa Hồ sơ cá nhân

Thông tin cá nhân

Họ và tên

Email (không thể thay đổi)

Số điện thoại

Vị trí / Chức vụ

Prof. Michael Brown

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Nhập số điện thoại

Nhập vị trí công tác

Thông tin học thuật

Học hàm

Học vị

-- Chọn Học hàm --

-- Chọn Học vị --

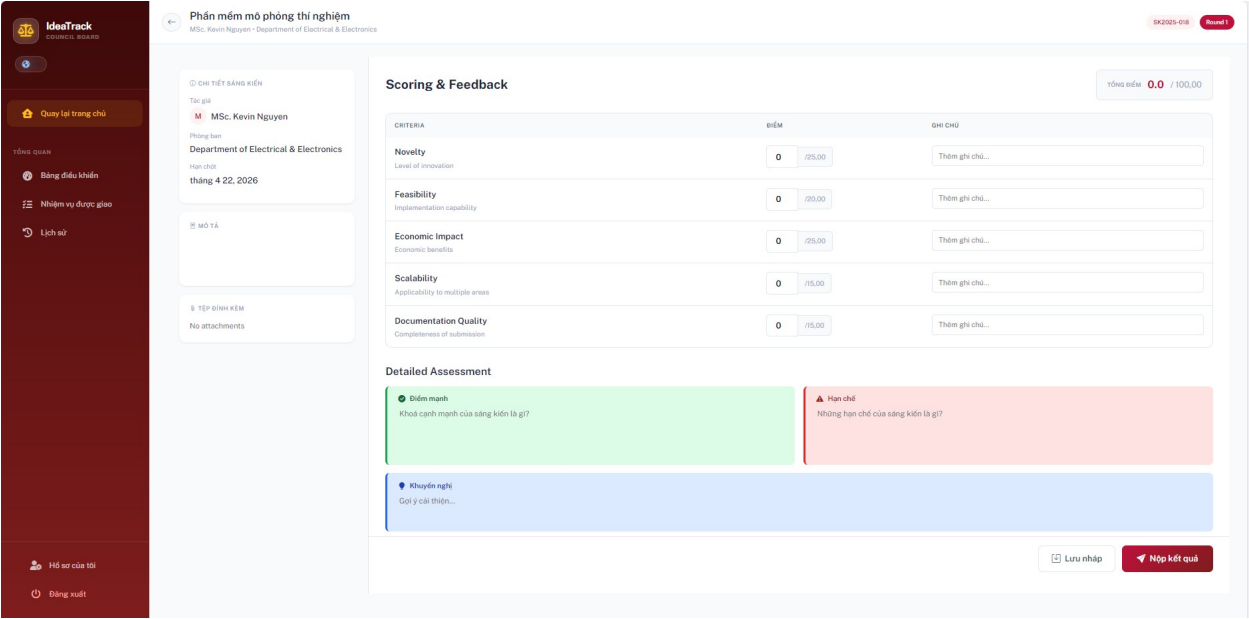
Picture Error: Reference source not found-7 Function faculty update profile

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
FA-PF-01	Update Profile — Valid Data (EP: valid partition)	Faculty user (Prof. Michael Brown) logged in.	1. Navigate to Profile Edit. 2. Set FullName = "Dr. Le B", Email = "leb@truong.edu.vn", Phone = "0901234567". 3. Click Save.	Profile updated successfully. No errors.	PASS
FA-PF-02	Update Profile — Invalid Email Format (EP: invalid email)	Faculty user logged in.	1. Navigate to Profile Edit. 2. Set Email = "leb-truong.edu.vn" (missing @). 3. Click Save.	Validation error: "Invalid email address."	PASS
FA-PF-03	Update Profile — Missing FullName (EP: invalid — empty)	Faculty user logged in.	1. Navigate to Profile Edit. 2. Clear FullName field. 3. Click Save.	Validation error: "Full name is required."	PASS
FA-PF-04	Update Profile — Phone with Letters (EP: invalid phone)	Faculty user logged in.	1. Navigate to Profile Edit. 2. Set Phone = "ABCDEF". 3. Click Save.	Validation error: "Invalid phone number format."	PASS
FA-PF-05	Update Profile — Phone Exactly 10 Digits	Faculty user logged in.	1. Set Phone = "0901234567" (10 digits). 2. Click Save.	Profile updated successfully.	PASS

	(BVA: valid boundary)				
FA-PF-06	Update Profile — Phone with 9 Digits (BVA: below minimum)	Faculty user logged in.	1. Set Phone = "090123456" (9 digits). 2. Click Save.	Validation error: "Phone number must be 10-11 digits."	PASS

Table Error: Reference source not found-8 Faculty profile testcases

3.2.3. Council module



Picture Error: Reference source not found-8 Function council review

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
CO-SC-01	Submit Scores — Maximum Valid Boundaries (BVA: upper bounds)	Council user logged in. Navigated to the Scoring page of an assigned initiative.	1. Enter max scores for all criteria: Novelty (25.00), Feasibility (20.00), Economic Impact (25.00), Scalability (15.00), Documentation Quality (15.00). 2. Fill in all Detailed Assessment text areas. 3. Click "Nộp kết quả" (Submit Result).	The "Tổng điểm" (Total Score) automatically calculates to exactly 100.00. Submission is successful and status updates.	PASS
CO-SC-	Input Scores — Invalid Upper	Council user logged in.	1. Enter 25.01 in the	System immediately displays a validation	PASS

02	Boundaries (BVA: max + 0.01)	Navigated to the Scoring page.	"Novelty" (max 25.00) field. 2. Click outside the field to trigger validation.	error (e.g., "Score must not exceed 25.00"). The input is rejected.	
CO-SC-03	Submit Scores — Minimum Valid Boundaries (BVA: lower bounds)	Council user logged in. Navigated to the Scoring page.	1. Enter 0 for all 5 criteria. 2. Fill in Detailed Assessment areas. 3. Click "Nộp kết quả".	The Total Score automatically calculates to 0.0. Submission is successful.	PASS
CO-SC-04	Input Scores — Invalid Lower Boundaries (BVA: negative values)	Council user logged in. Navigated to the Scoring page.	1. Enter -1 or -0.5 in any criteria field (e.g., Feasibility).	System prevents the input of negative numbers or displays a validation error: "Score cannot be less than 0".	PASS
CO-SC-05	Input Scores — Valid Decimal Formats (EP: decimal values)	Council user logged in. Navigated to the Scoring page.	1. Enter valid decimal scores (e.g., 12.5 or 12.50) into the criteria fields. 2. Check the Total Score calculation.	System accepts the decimal values correctly. Total score reflects the exact decimal addition.	PASS
CO-SC-06	Save Draft — Partial Data Entry (EP: incomplete form)	Council user logged in. Navigated to the Scoring page.	1. Enter a score for "Novelty" only. 2. Leave all other scores at 0 and assessment fields empty. 3. Click "Lưu nháp" (Save Draft).	System saves the current inputs without triggering mandatory field validations. User can return later to continue scoring.	PASS
CO-SC-07	Submit Scores — Missing Mandatory Assessments	Council user logged in. Navigated to the Scoring page.	1. Enter valid scores for all 5 criteria.	System blocks the submission and highlights the empty assessment text areas,	PASS

	(EP: empty text areas)		2. Leave the Detailed Assessment areas ("Điểm mạnh", "Hạn chế", "Khuyến nghị") completely blank. 3. Click "Nộp kết quả".	prompting the user to provide detailed feedback.	
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Table Error: Reference source not found-9 Council review testcases

[Return to Home](#)

OVERVIEW

- Dashboard
- Assigned Tasks
- History

[My Profile](#)
[Sign Out](#)

Evaluation History

4 Completed

Newest First

Apply Filter

INITIATIVE	CATEGORY	SCORE	DATE COMPLETED	STATUS	ACTION
AI-Powered Energy Optimization for Smart Campuses # SK-2026-0063	Software Initiative	85 / 100	Apr 07, 2026	Completed	View Details
Smart Library Management System # SK2025-001	Cơ sở vật chất	90 / 100	Apr 06, 2026	Completed	View Details
Integration Test Initiative 2026 v2 # SK-2026-0060	Hardware/IoT Initiative	75 / 100	Apr 06, 2026	Completed	View Details
itest # SK-2026-0055	ctest	150 / 100	Mar 31, 2026	Completed	View Details

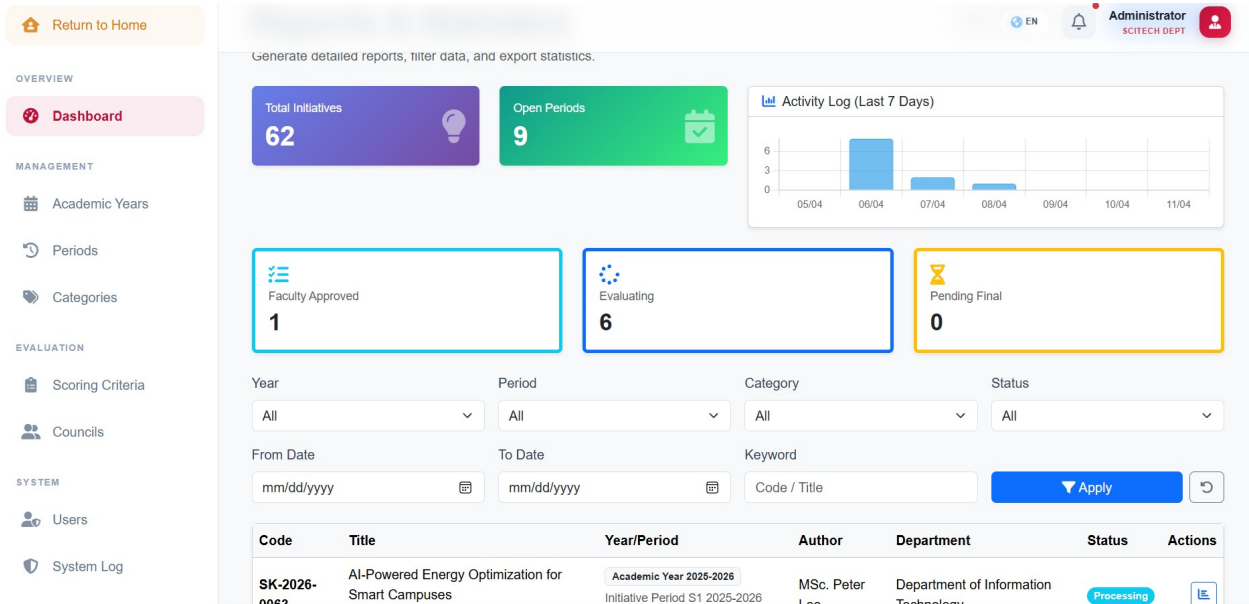
Picture Error: Reference source not found-9 Function council history

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
CO-AT-01	View Assigned Tasks (EP: valid view)	Council user (council1) logged in. Has assigned initiatives.	1. Navigate to Dashboard. 2. Click "Assigned Tasks" in the sidebar.	Assigned initiatives list is displayed, sorted by deadline. Shows SK2025-004 "Automatic Grading Software."	PASS
CO-HI-01	View Evaluation History — Empty (EP: no history)	Council user logged in. No submissions completed yet.	1. Navigate to Dashboard. 2. Click "History" in the sidebar.	Message: "No evaluation history found. Complete your first initiative to see it here!"	PASS
CO-HI-02	Search History with Non-Existent Keyword	Council user logged in. History page is displayed.	1. Navigate to History. 2. Enter keyword = "XYZABC999NOTFOUND". 3. Click Search.	Message: "No evaluation history found."	PASS

	(EP: no match)				
CO-AT-02	Filter Assigned Tasks — All Tab (EP: valid filter)	Council user logged in. Has mixed status initiatives.	1. Navigate to Assigned Tasks. 2. Click "All" tab.	All assigned initiatives are displayed regardless of status.	PASS
CO-AT-03	Filter Assigned Tasks — Completed Tab (EP: valid filter)	Council user logged in.	1. Navigate to Assigned Tasks. 2. Click "Completed" tab.	Only completed initiatives are displayed. If none, appropriate empty message shown.	PASS
CO-AT-04	Search Assigned Tasks by Code (EP: code search)	Council user logged in. Initiative SK2025-004 assigned.	1. Navigate to Assigned Tasks. 2. Enter "SK2025-004" in search bar.	Only initiative SK2025-004 is displayed.	PASS
CO-AT-05	Search Assigned Tasks — Empty Keyword (BVA: empty)	Council user logged in.	1. Navigate to Assigned Tasks. 2. Leave search empty. 3. Click Search.	All assigned initiatives are displayed (no filter).	PASS

Table Error: Reference source not found-10 Council history activity testcases

3.2.4. Scitech module



Picture Error: Reference source not found-10 Function scitech dashboard

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
ST-FI-01	Filter by Status = "Evaluating" (EP: valid status	SciTech user logged in. Dashboard is	1. In "Status" dropdown, select "Evaluating".	URL updates to status=Evaluating. List shows only "Evaluating"	PASS

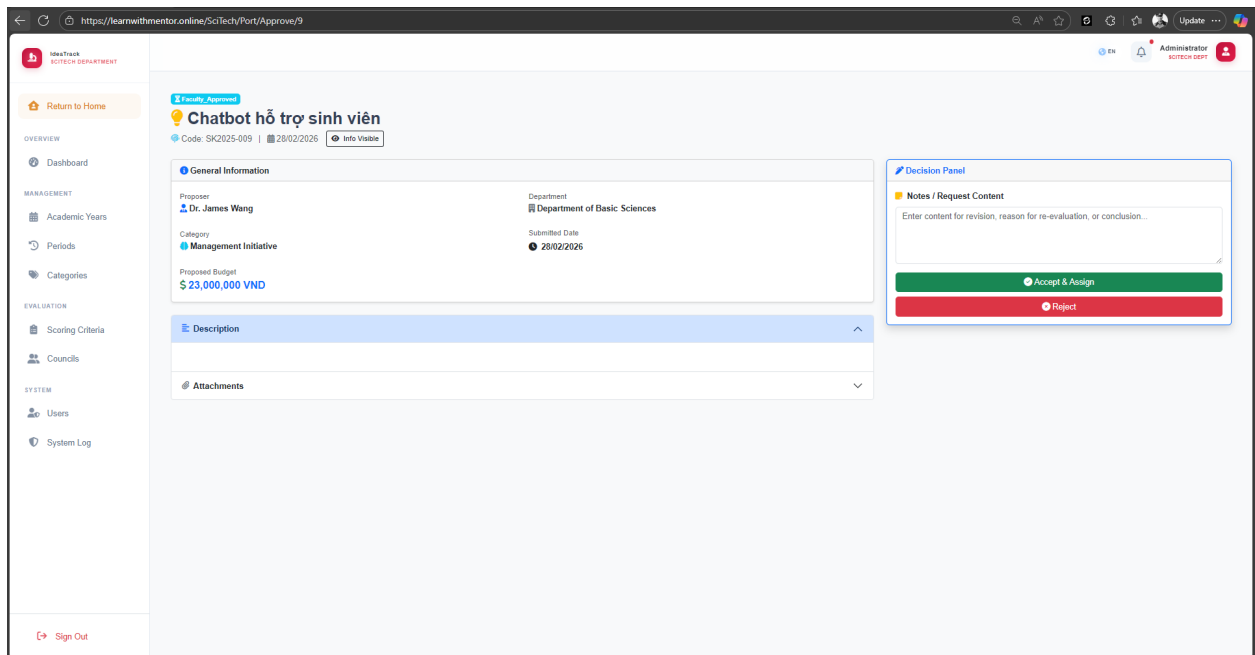
	filter)	displayed.	2. Click "Apply".	initiatives.	
ST-FI-02	Search by Keyword "IoT" (EP: valid keyword match)	SciTech user logged in. Dashboard is displayed.	1. Enter "IoT" in Keyword field. 2. Click "Apply".	Returns 1 result: "IoT Environment Monitoring System" (SK2025-011).	PASS
ST-FI-03	Filter by Date Range — Today Only (BVA: FromDate = ToDate = Today)	SciTech user logged in.	1. Set FromDate = today. 2. Set ToDate = today. 3. Click "Apply".	Only today's submissions are listed.	PASS
ST-FI-04	Filter with FromDate > ToDate (BVA: invalid range)	SciTech user logged in.	1. Set FromDate = "2026-04-15". 2. Set ToDate = "2026-04-10". 3. Click "Apply".	Validation error or empty results returned.	PASS
ST-FI-05	Filter by All Statuses (EP: no filter = all)	SciTech user logged in.	1. Select Status = "All". 2. Click "Apply".	All initiatives regardless of status are displayed.	PASS
ST-FI-06	Search with Empty Keyword (BVA: empty string)	SciTech user logged in.	1. Leave Keyword field empty. 2. Click "Apply".	All initiatives are displayed (no keyword filter).	PASS
ST-FI-07	Search with Very Long Keyword (BVA: 500+ chars)	SciTech user logged in.	1. Enter 500+ character string in Keyword. 2. Click "Apply".	System handles gracefully — no crash. Returns empty result or truncates input.	PASS

Table Error: Reference source not found-11 Scitech dashboard testcases

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
ST-EX-01	Export Data to Excel (EP: valid export)	SciTech user logged in. Dashboard is displayed.	1. Click "Export Data" button.	File "BaoCaoHoSo.xlsx" is downloaded. Contains all 52 initiatives with columns: Code, Title, Creator, Status, etc.	PASS
ST-EX-02	Print Report as PDF (EP: valid PDF generation)	SciTech user logged in. Dashboard is displayed.	1. Click "Print Report (PDF)" button.	File "BaoCaoHoSo.pdf" is downloaded. Content is well-formatted with complete initiative list.	PASS
ST-EX-03	Export Data with Active Filters (EP: filtered)	SciTech user logged in. Status filter =	1. Set filter Status = "Evaluating". 2. Click "Export	Exported file contains only initiatives with "Evaluating" status.	PASS

	export)	"Evaluating".	Data".		
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Table Error: Reference source not found-12 Scitech report testcases

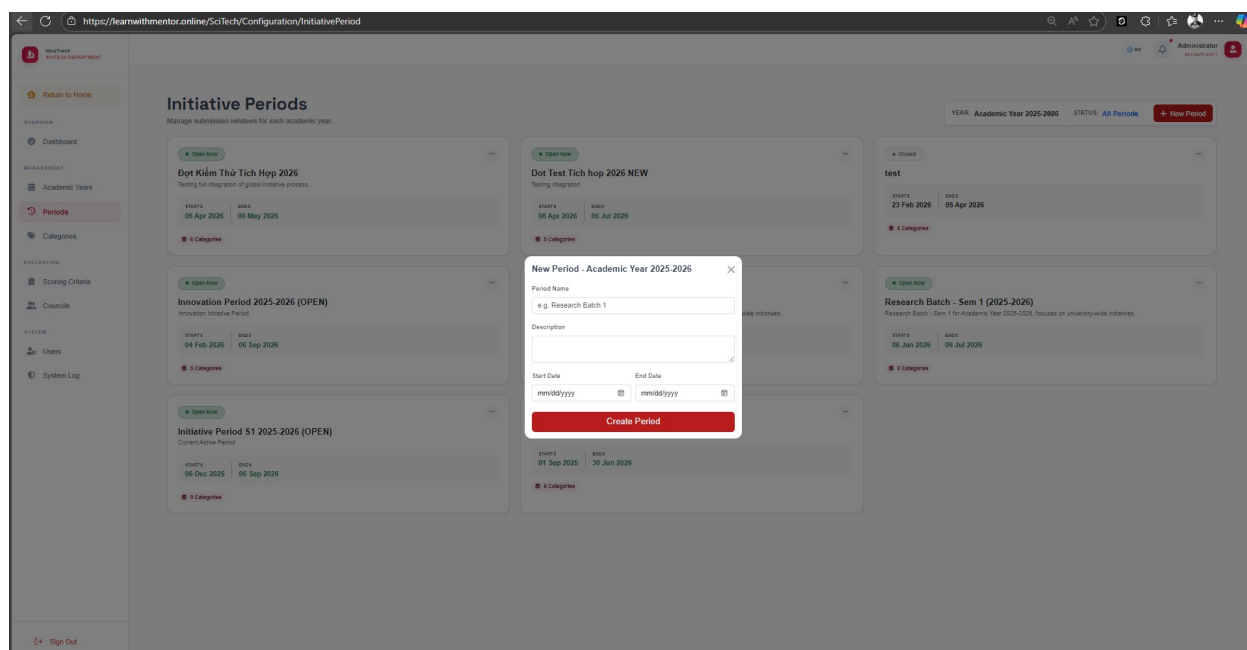


Picture Error: Reference source not found-11 Function scitech review initiative

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
ST-INIT-01	Approve New Initiative from Faculty (EP: Status = Faculty_Approved)	SciTech user logged in. Initiative status = Faculty_Approved.	1. Navigate to Port > Approve. 2. Open Faculty_Approved initiative. 3. Click "Accept & Assign".	Status updates to Evaluating. Council members and scoring template are auto-assigned.	PASS
ST-INIT-02	Reject Initiative at SciTech Stage (EP: Faculty_Approved → Rejected)	SciTech user logged in. Initiative status = Faculty_Approved.	1. Navigate to Port > Approve. 2. Open initiative. 3. Enter rejection note. 4. Click "Reject".	Status updates to Rejected. User is notified via RevisionRequest log.	PASS
ST-INIT-03	Request New Evaluation Round (EP: Round 1 → Round 2)	SciTech user logged in. Initiative CurrentRound = 1. Status = Evaluating.	1. Navigate to Port > Result. 2. Select initiative. 3. Click "Request New Round".	Status remains Evaluating or Re_Evaluating. CurrentRound incremented to 2.	PASS
ST-INIT-04	Make Final Approve Decision (EP: Status = Faculty_Approved)	SciTech user logged in. Initiative status = Faculty_Approved.	1. Navigate to Port > Result. 2. Select initiative.	Status updates to Processing. Final report is generated.	PASS

	Pending_Final)	Pending_Final.	3. Click "Approve" (Final Decision).		
ST-INIT-05	Toggle Personal Info Visibility (EP: Toggle Anonymization)	SciTech user logged in. Initiative exists.	1. Navigate to Port > Result. 2. Toggle HidePersonalInfo = True. 3. View evaluation.	Author names are hidden in evaluation view and PDF export.	PASS
ST-INIT-06	Filter Initiatives by Date Range (BVA: Today only)	SciTech user logged in.	1. Set FromDate = Today, ToDate = Today. 2. Click "Apply".	Only today's submissions are listed.	PASS
ST-INIT-07	Approve Initiative Without Notes (BVA: empty notes)	SciTech user logged in. Initiative status = Faculty_Approved.	1. Navigate to Port > Approve. 2. Leave "Notes" empty. 3. Click "Accept & Assign".	Initiative is accepted successfully even without notes.	PASS
ST-INIT-08	Reject Initiative Without Notes (BVA: empty notes)	SciTech user logged in. Initiative status = Faculty_Approved.	1. Navigate to Port > Approve. 2. Leave "Notes" empty. 3. Click "Reject".	Validation error if notes are required, or rejection proceeds with empty notes.	PASS

Table Error: Reference source not found-13 Scitech review initiative testcases



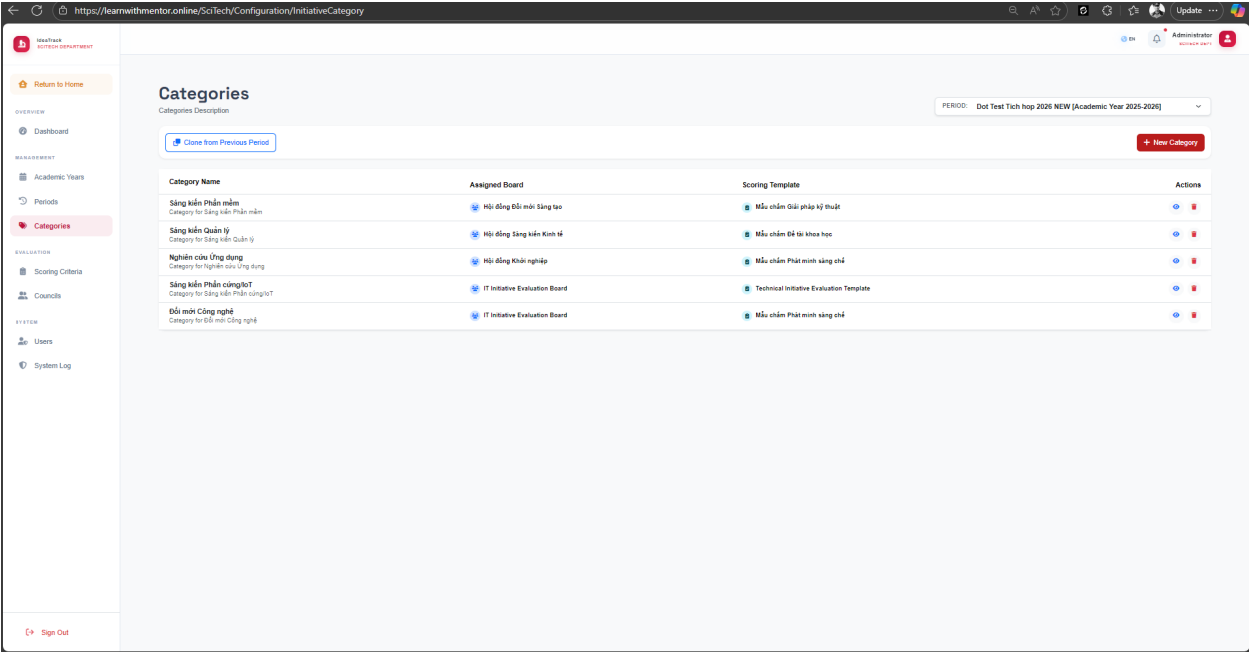
Picture Error: Reference source not found-12 Function scitech create period

ID	Title	Pre-conditions	Steps	Expected / Actual	Status
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				Results	
ST-PER-01	Create Period with Valid Dates (EP: StartDate < EndDate)	SciTech user logged in. Academic Year exists.	1. Navigate to Periods. 2. Click "+ New Period". 3. Enter name, StartDate = "2026-01-01", EndDate = "2026-06-30". 4. Click "Create Period".	Period is created and linked to the correct Academic Year.	PASS
ST-PER-02	Create Period with Invalid Date Range (BVA: StartDate = EndDate + 1 day)	SciTech user logged in.	1. Navigate to Periods. 2. Click "+ New Period". 3. Set StartDate = "2026-07-01", EndDate = "2026-06-30". 4. Click "Create Period".	Validation error: "End date must be after Start date."	PASS
ST-PER-03	Create Period Currently "Open" (BVA: StartDate ≤ Today ≤ EndDate)	SciTech user logged in.	1. Create period with StartDate in the past, EndDate in the future. 2. Click "Create Period".	Period is created. Badge shows "Open" on dashboard.	PASS
ST-PER-04	Create Period "Upcoming" (BVA: StartDate > Today)	SciTech user logged in.	1. Create period with StartDate in the future. 2. Click "Create Period".	Period is created. Badge shows "Closed" (Upcoming).	PASS
ST-PER-05	Clone Categories from Previous Period (EP: valid clone)	SciTech user logged in. Previous period with categories exists.	1. Navigate to Periods. 2. Click "+ New Period". 3. Fill dates. 4. Select CloneFromPeriodId. 5. Click "Create Period".	New period created with all categories duplicated from the source period.	PASS
ST-PER-06	Create Period with StartDate = EndDate (BVA: same day)	SciTech user logged in.	1. Click "+ New Period". 2. Set StartDate = EndDate = "2026-05-01". 3. Click Create.	Validation error or period created with 1-day duration.	PASS
ST-PER-07	Create Period with Empty Name (EP: invalid — blank)	SciTech user logged in.	1. Click "+ New Period". 2. Leave Period Name blank. 3. Set valid dates. 4. Click Create.	Validation error: "Period name is required."	PASS
ST-PER-08	Create Academic Year — Valid (EP: valid)	SciTech user logged in. Academic Years page displayed.	1. Navigate to Academic Years. 2. Click "+ New Academic Year". 3. Enter "Academic Year	Academic Year is created and appears in the list.	PASS

			2026-2027". 4. Click "Create Year".		
ST-PER-09	Create Academic Year — Duplicate Name (EP: invalid — duplicate)	SciTech user logged in. "Academic Year 2025-2026" already exists.	1. Click "+ New Academic Year". 2. Enter "Academic Year 2025-2026". 3. Click "Create Year".	Validation error: "Academic Year already exists."	PASS
ST-PER-10	Create Academic Year — Empty Name (BVA: blank)	SciTech user logged in.	1. Click "+ New Academic Year". 2. Leave Display Name empty. 3. Click "Create Year".	Validation error: "Display name is required."	PASS

Table Error: Reference source not found-14 Scitech period CRUD testcases



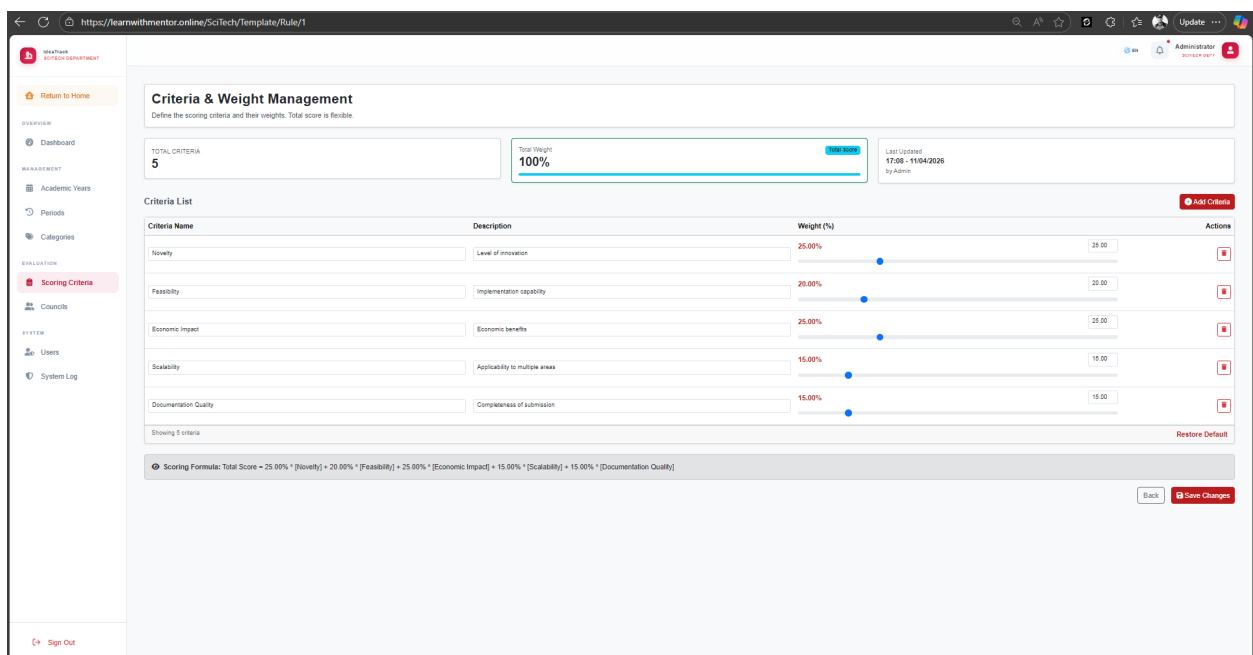
Picture Error: Reference source not found-13 Function scitech manage categories

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
ST-CAT-01	View Categories for a Period (EP: valid view)	SciTech admin logged in. Period with categories exists.	1. Navigate to Categories. 2. Select Period from dropdown.	Category list is displayed with columns: Category Name, Assigned Board, Scoring Template, Actions.	PASS
ST-CAT-02	Create New Category (EP: valid)	SciTech admin logged in.	1. Navigate to Categories. 2. Click "+ New	Category is created and appears in the list with correct Assigned Board	PASS

	creation)	Period is selected.	Category". 3. Enter category name, assign board and template. 4. Click Create.	and Scoring Template.	
ST-CAT-03	Create Category — Empty Name (BVA: blank)	SciTech admin logged in. Period is selected.	1. Click "+ New Category". 2. Leave Category Name blank. 3. Click Create.	Validation error: "Category name is required."	PASS
ST-CAT-04	Create Category — Duplicate Name (EP: duplicate within period)	SciTech admin logged in. Category "Software Initiative" exists in current period.	1. Click "+ New Category". 2. Enter name = "Software Initiative". 3. Click Create.	Validation error: "Category already exists in this period."	PASS
ST-CAT-05	Clone Categories from Previous Period (EP: valid clone)	SciTech admin logged in. Previous period has categories.	1. Navigate to Categories. 2. Click "Clone from Previous Period". 3. Select source period. 4. Confirm.	All categories from the source period are cloned into the current period with their assigned boards and templates.	PASS
ST-CAT-06	Delete Category (EP: valid deletion)	SciTech admin logged in. Category exists in the selected period.	1. Navigate to Categories. 2. Click delete icon (red trash) next to a category. 3. Confirm deletion.	Category is removed from the list.	PASS
ST-CAT-07	View Category Details (EP: valid view)	SciTech admin logged in. Category exists.	1. Navigate to Categories. 2. Click view icon (eye) next to a category.	Category detail page is displayed showing assigned board members, scoring template, and linked initiatives.	PASS
ST-CAT-08	Edit Category — Change Assigned Board (EP: valid edit)	SciTech admin logged in. Category exists.	1. Navigate to Categories. 2. Click edit icon next to a category. 3. Change Assigned Board.	Category is updated with new Assigned Board. Change is reflected in the list.	PASS

			4. Click Save.		
ST-CAT-09	Edit Category — Change Scoring Template (EP: valid edit)	SciTech admin logged in. Category exists. Multiple templates available.	1. Click edit icon next to a category. 2. Change Scoring Template. 3. Click Save.	Category is updated with new Scoring Template.	PASS
ST-CAT-10	Switch Period in Dropdown (EP: valid filter)	SciTech admin logged in. Multiple periods exist.	1. Navigate to Categories. 2. Change Period dropdown from one period to another.	Category list updates to show categories for the newly selected period.	PASS

Table Error: Reference source not found-15 Scitech categories CRUD testcases



Picture Error: Reference source not found-14 Function scitech update scoring criteria

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
ST-TMP-01	Create Scoring Template — Valid (EP: valid string)	SciTech user logged in. Template page displayed.	1. Navigate to Scoring Criteria. 2. Click "+ Create Template". 3. Enter TemplateName. 4. Click Create.	Template is created with status Inactive by default.	PASS
ST-TMP-02	Add Criteria with Boundary Scores (BVA: MaxScore = 0 and 500+)	SciTech user logged in. Template exists.	1. Navigate to Template Rules. 2. Add criteria with MaxScore = 0.	System accepts values ≥ 0 . Values above 100 allowed for complex	PASS

			3. Add criteria with MaxScore = 500.	rubrics.	
ST-TMP-03	Activate Template (EP: Toggle IsActive)	SciTech user logged in. Inactive template exists.	1. Navigate to Template Index. 2. Click "Edit" on inactive template. 3. Toggle IsActive to Active. 4. Save.	Template status changes to Active. Can be linked to Categories.	PASS
ST-TMP-04	Create Template — Empty Name (BVA: blank)	SciTech user logged in.	1. Click "+ Create Template". 2. Leave TemplateName blank. 3. Click Create.	Validation error: "Template name is required."	PASS
ST-TMP-05	Create Template — Duplicate Name (EP: duplicate)	SciTech user logged in. Template with same name exists.	1. Click "+ Create Template". 2. Enter existing template name. 3. Click Create.	Validation error or template created (if duplicates allowed).	PASS
ST-TMP-06	Add Criteria — Weight Sum > 100% (BVA: over 100%)	SciTech user logged in. Template has criteria totaling 100%.	1. Add new criteria with Weight = 10%. 2. Click "Save Changes".	Validation error: "Total weight exceeds 100%." Or system allows flexible weights.	PASS
ST-TMP-07	Add Criteria — Weight = 0% (BVA: lower bound)	SciTech user logged in. Template exists.	1. Add new criteria with Weight = 0%. 2. Click "Save Changes".	Criteria is added with 0% weight (valid edge case).	PASS
ST-TMP-08	Add Criteria — Negative MaxScore (EP: invalid)	SciTech user logged in. Template exists.	1. Add criteria with MaxScore = -10. 2. Click Save.	Validation error: "Max score must be non-negative."	PASS
ST-TMP-09	Delete Template (EP: delete action)	SciTech user logged in. Inactive template exists.	1. Click "Edit" on template. 2. Click "Delete". 3. Confirm deletion.	Template is deleted. Removed from template list.	PASS

Table Error: Reference source not found-16 Scitech criteria CRUD testcases

User Management
User Management Description

Overview: Total Users: 27, Active Now: 100%, Recent Logins: 0, Pending Approval: 0.

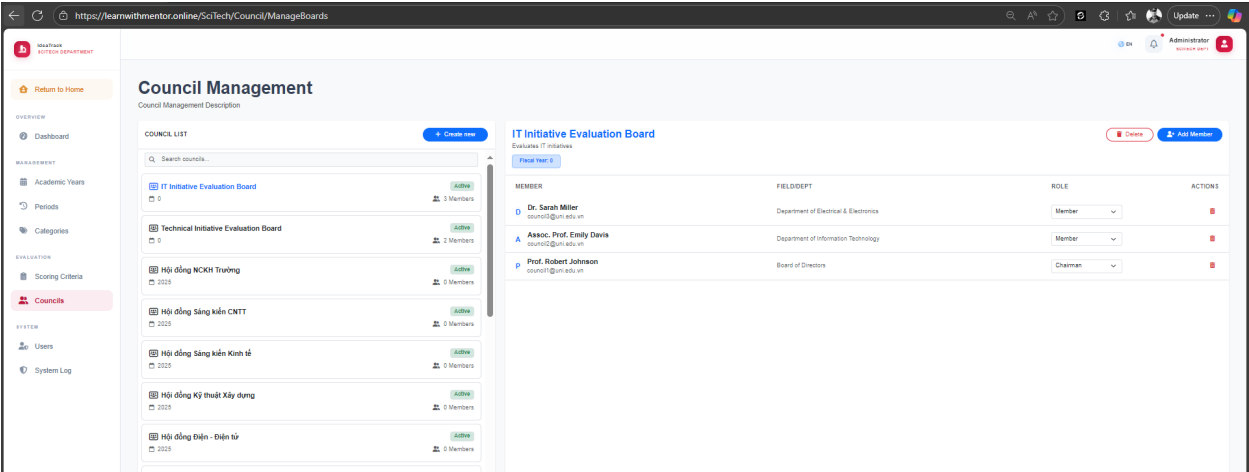
Full Name	Role	Status	Last Login	Login Count	Actions
Administrator admin@uni.edu.vn	Admin	Active	Never	0	[Edit] [Delete]
John Smith scitech1@uni.edu.vn	SciTech	Active	Never	2	[Edit] [Delete]
Jane Doe scitech2@uni.edu.vn	SciTech	Active	Never	0	[Edit] [Delete]
Prof. Michael Brown reader.crit@uni.edu.vn	FacultyLeader	Active	Never	0	[Edit] [Delete]
Dr. David Wilson reader.ot@uni.edu.vn	FacultyLeader	Active	Never	4	[Edit] [Delete]
Prof. Robert Johnson council1@uni.edu.vn	CouncilMember	Active	Never	0	[Edit] [Delete]
Assoc. Prof. Emily Davis council2@uni.edu.vn	CouncilMember	Active	Never	1	[Edit] [Delete]
Dr. Sarah Miller council3@uni.edu.vn	CouncilMember	Active	Never	4	[Edit] [Delete]
MSc. Peter Lee author1@uni.edu.vn	Lecturer	Active	Never	0	[Edit] [Delete]
MSc. Mary Chen author2@uni.edu.vn	Lecturer	Active	Never	3	[Edit] [Delete]

Picture Error: Reference source not found-15 Function scitech manage users

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
ST-USER-01	Create User with Specific Role (EP: valid roles)	SciTech admin logged in. User management page displayed.	1. Navigate to Users. 2. Click "+ New User". 3. Enter user details with Role = "Council". 4. Click Create.	User is created and assigned correct ASP.NET Identity Role.	PASS
ST-USER-02	Lock Active User Account (EP: IsActive = False)	SciTech admin logged in. Active user exists.	1. Navigate to Users. 2. Find active user. 3. Click Lock/Deactivate.	User status changes to Inactive. User can no longer log in.	PASS
ST-USER-03	Export User List to Excel (EP: filtered export)	SciTech admin logged in. Users with various roles exist.	1. Navigate to Users. 2. Filter by Role = "Council". 3. Click "Export Report".	Excel file downloaded containing only Council users.	PASS
ST-USER-04	Create User — Invalid Email (EP: invalid email)	SciTech admin logged in.	1. Click "+ New User". 2. Enter email = "notanemail". 3. Click Create.	Validation error: "Invalid email address."	PASS
ST-USER-05	Create User Without Role (EP: missing required field)	SciTech admin logged in.	1. Click "+ New User". 2. Enter user details. 3. Do NOT select a role. 4. Click Create.	Validation error: "Role is required."	PASS
ST-USER-06	Unlock Locked User	SciTech admin logged in.	1. Find locked user. 2. Click Unlock/Activate.	User status changes to Active. User can	PASS

6	Account (EP: IsActive = True)	Locked user exists.		log in again.	
ST- USER-0 7	Search Users by Name (EP: valid search)	SciTech admin logged in.	1. Enter "Peter" in search bar. 2. Click Filter.	Only users with "Peter" in their name are displayed.	PASS
ST- USER-0 8	Search Users by Email (EP: valid search)	SciTech admin logged in.	1. Enter "council" in search bar. 2. Click Filter.	Users with "council" in their name or email are displayed.	PASS
ST- USER-0 9	Filter Users by Role and Status (EP: combined filter)	SciTech admin logged in.	1. Set Role = "Lecturer". Set Status = "Active". 2. Click Filter.	Only active Lecturer users are displayed.	PASS
ST- USER-1 0	Create User — Duplicate Email (EP: duplicate)	SciTech admin logged in. User "author1@uni.edu.vn" exists.	1. Click "+ New User". 2. Enter email = "author1@uni.edu.vn". 3. Click Create.	Validation error: "Email already exists."	PASS

Table Error: Reference source not found-17 Scitech users CRUD testcases



Picture Error: Reference source not found-16 Function scitech council management

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
ST- CL-01	Create New Council Board (EP: valid creation)	SciTech admin logged in. Council management page	1. Navigate to Councils. 2. Click "+ Create new".	Council board is created and appears in the council list.	PASS

		displayed.	3. Enter council name. 4. Click Create.		
ST-CL-02	Add Member to Council Board (EP: valid addition)	SciTech admin logged in. Council board exists.	1. Select a council board. 2. Click "+ Add Member". 3. Select user and role. 4. Click Save.	Member is added to the council. Role is correctly assigned.	PASS
ST-CL-03	Remove Member from Council Board (EP: valid removal)	SciTech admin logged in. Council board has members.	1. Select a council board. 2. Click delete icon next to a member. 3. Confirm removal.	Member is removed from the council board.	PASS
ST-CL-04	Delete Council Board (EP: valid deletion)	SciTech admin logged in. Council board exists.	1. Select a council board. 2. Click "Delete". 3. Confirm deletion.	Council board is deleted and removed from the list.	PASS
ST-CL-05	Create Council Board — Empty Name (BVA: blank)	SciTech admin logged in.	1. Click "+ Create new". 2. Leave name blank. 3. Click Create.	Validation error: "Council name is required."	PASS
ST-CL-06	Search Council Boards (EP: valid search)	SciTech admin logged in. Multiple councils exist.	1. Enter "IT" in search bar.	Only councils matching "IT" are displayed.	PASS

Table Error: Reference source not found-18 Scitech council CRUD testcases

IdeaTrack

NOTECHE DEPARTMENT

EN

Administrator

NOTECHE DEPT

Return to Home

OVERVIEW

Dashboard

Academic Years

Periods

Categories

EVALUATION

Scoring Criteria

Councils

SYSTEM

Users

System Log

Audit Log

Statistics

Hiện thị

▼

đóng

Tìm kiếm

Người Dùng	Hành Động	Bảng	ID	Thời Gian	IP	Chi Tiết
Minh	Reverse	Initiatives	1	08/04/2026 19:48	:1	
MSc. Peter Lee	Update	Initiatives	63	07/04/2026 13:03	:1	
MSc. Peter Lee	Create	Initiatives	63	07/04/2026 12:31	:1	
Minh	Approve	Initiatives	1	06/04/2026 20:26	:1	
MSc. Peter Lee	Update	Initiatives	1	06/04/2026 20:10	:1	
MSc. Peter Lee	Create	Initiatives	61	06/04/2026 20:04	:1	
MSc. Peter Lee	Create	Initiatives	60	06/04/2026 19:47	:1	
MSc. Peter Lee	Update	Initiatives	59	06/04/2026 19:45	:1	
MSc. Peter Lee	Create	Initiatives	59	06/04/2026 19:39	:1	
MSc. Peter Lee	Update	Initiatives	58	06/04/2026 17:49	:1	
MSc. Peter Lee	Create	Initiatives	58	06/04/2026 17:48	:1	
MSc. Peter Lee	Create	Initiatives	56	01/04/2026 10:37	:1	
John Smith	Login	EvaluationDetails	8	31/03/2026 22:43	192.168.1.80	
Prof. Michael Brown	Approve	EvaluationDetails	2	31/03/2026 22:15	192.168.1.104	
MSc. Peter Lee	Create	Initiatives	55	31/03/2026 21:56	:1	

Tất cả

Tất cả

Tất cả

ID

Thời Gian

IP

Chi Tiết

Đang hiển thị 1 đến 15 của 113 mục

Trước

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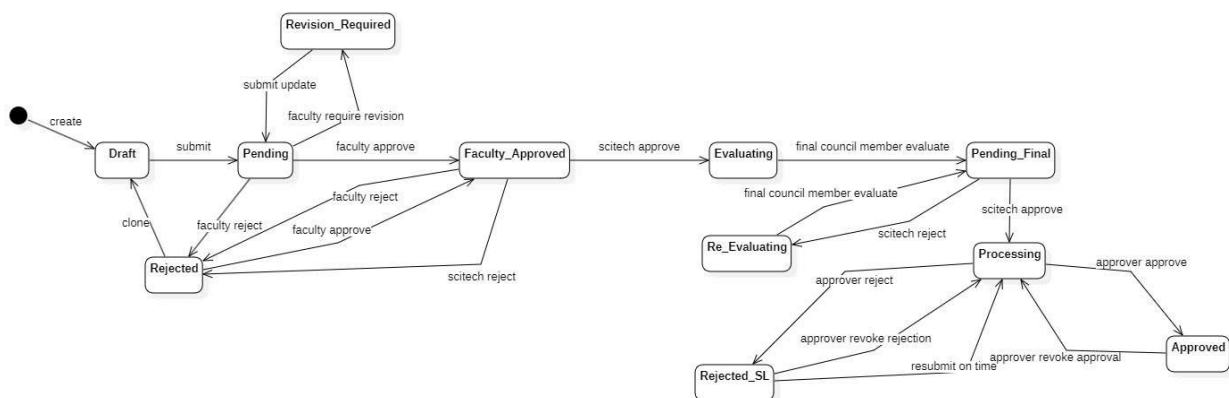
Sau

Picture Error: Reference source not found-17 Function audit logs

ID	Title	Pre-conditions	Steps	Expected / Actual Results	Status
ST-LOG-01	View Audit Log (EP: valid view)	SciTech admin logged in. System actions have been performed.	1. Navigate to System Log.	Audit log is displayed with columns: User, Action, Table, ID, Timestamp, IP, Details.	PASS
ST-LOG-02	Filter Audit Log by Action Type (EP: valid filter)	SciTech admin logged in.	1. Navigate to System Log. 2. Filter by Action = "Create".	Only "Create" actions are displayed.	PASS
ST-LOG-03	Search Audit Log by User (EP: valid search)	SciTech admin logged in.	1. Navigate to System Log. 2. Enter "Peter Lee" in search bar.	Only logs for "MSc. Peter Lee" are displayed.	PASS
ST-LOG-04	Audit Log Pagination (EP: multi-page)	SciTech admin logged in. More than 15 log entries exist.	1. Navigate to System Log. 2. Verify pagination. 3. Click page 2.	Page 2 displays entries 16-30. Pagination works correctly.	PASS

Table Error: Reference source not found-19 Audit logs testcases

3.2.5. Initiative state flow



Picture Error: Reference source not found-18 Initiative statechart diagram

TC ID	Title	Pre-conditions	Steps	Expected / Actual Results
TC-E2E-001	End-to-End: Complete approval flow from Draft to Approved	Users with roles Author, Faculty, SciTech, Council Member, and Approver are available and logged	1. Author: Create a new initiative → status = Draft. 2. Author: Submit the initiative → status = Pending. 3. Faculty: Approve the initiative → status =	- Initiative successfully transitions: Draft → Pending → Faculty_Approved → Evaluating → Pending_Final →

		in.	Faculty_Approved. 4. SciTech: Approve the initiative → status = Evaluating. 5. Council Members: Evaluate (final member completes) → status = Pending_Final. 6. SciTech: Approve final evaluation → status = Processing. 7. Approver: Approve the initiative → status = Approved.	Processing → Approved. - Each transition is recorded in the audit log. - All stakeholders are notified at each transition. - Final status is Approved.
TC-E2E-002	End-to-End: Faculty requests revision, then full approval	Users with roles Author, Faculty, SciTech, Council Member, and Approver are available.	1. Author: Create and submit an initiative → status = Pending. 2. Faculty: Require revision → status = Revision_Required. 3. Author: Submit updated initiative → status = Pending. 4. Faculty: Approve the revised initiative → status = Faculty_Approved. 5. SciTech: Approve → status = Evaluating. 6. Council Members: Evaluate → status = Pending_Final. 7. SciTech: Approve → status = Processing. 8. Approver: Approve → status = Approved.	- Initiative follows: Draft → Pending → Revision_Required → Pending → Faculty_Approved → Evaluating → Pending_Final → Processing → Approved. - Revision history is maintained. - Both the original and revised content are accessible.
TC-E2E-003	End-to-End: Faculty rejects, Author clones, then full approval	Users with roles Author and Faculty are available.	1. Author: Create and submit an initiative → status = Pending. 2. Faculty: Reject the initiative → status = Rejected. 3. Author: Clone the rejected initiative → new initiative in Draft status. 4. Author: Modify and submit the cloned initiative → status = Pending.	- Original initiative remains in Rejected status. - Cloned initiative progresses through the full approval flow. - Both initiatives have separate audit trails. - Final cloned initiative status is Approved.

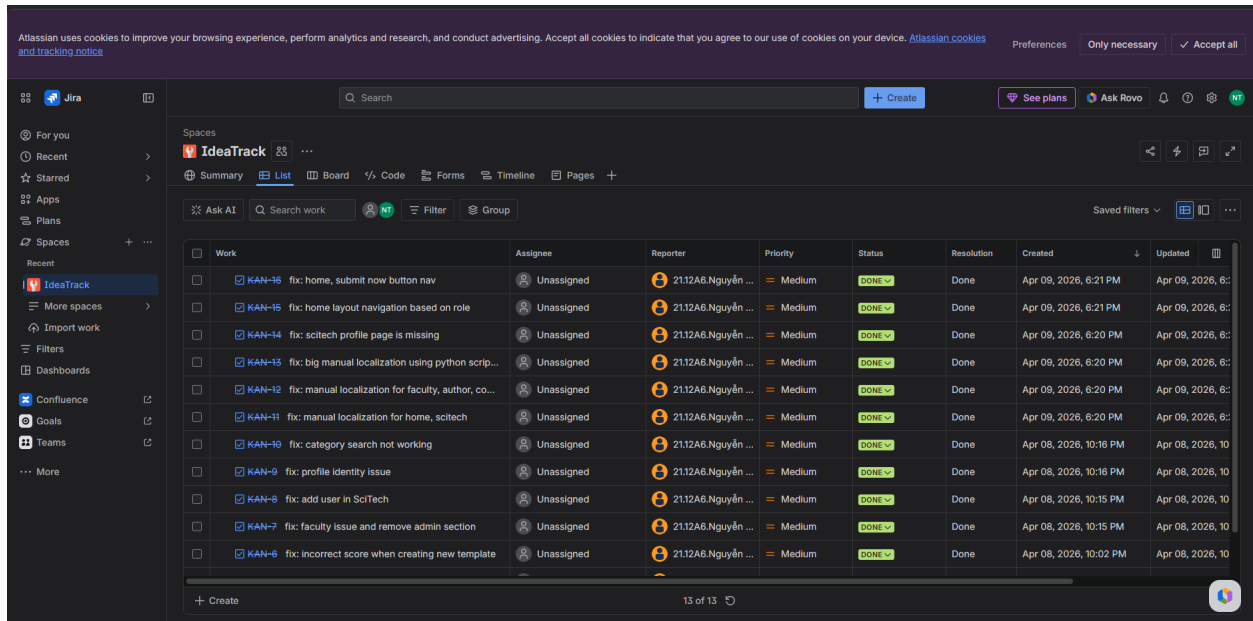
			5. Faculty: Approve the resubmitted initiative → status = Faculty_Aproved. 6. Continue through the remaining approval flow to Approved.	
TC-E2E-004	End-to-End: Approver rejects, then revokes rejection and approves	An initiative exists in Processing status. User is logged in with Approver role.	1. Approver: Reject the initiative → status = Rejected_SL. 2. Approver: Revoke the rejection → status = Processing. 3. Approver: Approve the initiative → status = Approved.	- Initiative transitions: Processing → Rejected_SL → Processing → Approved. - All three state changes are recorded in the audit log. - Final status is Approved.
TC-E2E-005	End-to-End: Approver approves, revokes approval, then rejects	An initiative exists in Processing status. User is logged in with Approver role.	1. Approver: Approve the initiative → status = Approved. 2. Approver: Revoke approval → status = Processing. 3. Approver: Reject the initiative → status = Rejected_SL.	- Initiative transitions: Processing → Approved → Processing → Rejected_SL. - Revocation reason is recorded. - Final status is Rejected_SL.
TC-E2E-006	End-to-End: Council triggers re-evaluation, then full approval	An initiative exists in Evaluating status. Council Members and Approver are available.	1. Council Members: Evaluate, triggering re-evaluation → status = Re_Evaluating. 2. Council Members: Complete re-evaluation → status = Processing. 3. Approver: Approve the initiative → status = Approved.	- Initiative transitions: Evaluating → Re_Evaluating → Processing → Approved. - Both original and re-evaluation scores are recorded. - Final status is Approved.

Table Error: Reference source not found-20 Initiative stateflow testcases

CHAPTER 4. Defect Report & Metrics

4.1. Defect Log

Below is the list of defects identified and fixed during previous testing cycles, extracted from the IdeaTrack project's Jira issue tracking system:



Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated
☐ KAN-16 fix: home, submit now button nav	Unassigned	21.12A6.Nguyễn ...	Medium	DONE	Done	Apr 09, 2026, 6:21 PM	Apr 09, 2026, 6:21 PM
☐ KAN-15 fix: home layout navigation based on role	Unassigned	21.12A6.Nguyễn ...	Medium	DONE	Done	Apr 09, 2026, 6:21 PM	Apr 09, 2026, 6:21 PM
☐ KAN-14 fix: scitech profile page is missing	Unassigned	21.12A6.Nguyễn ...	Medium	DONE	Done	Apr 09, 2026, 6:20 PM	Apr 09, 2026, 6:20 PM
☐ KAN-13 fix: big manual localization using python scrip...	Unassigned	21.12A6.Nguyễn ...	Medium	DONE	Done	Apr 09, 2026, 6:20 PM	Apr 09, 2026, 6:20 PM
☐ KAN-12 fix: manual localization for faculty, author, co...	Unassigned	21.12A6.Nguyễn ...	Medium	DONE	Done	Apr 09, 2026, 6:20 PM	Apr 09, 2026, 6:20 PM
☐ KAN-11 fix: manual localization for home, scitech	Unassigned	21.12A6.Nguyễn ...	Medium	DONE	Done	Apr 09, 2026, 6:20 PM	Apr 09, 2026, 6:20 PM
☐ KAN-10 fix: category search not working	Unassigned	21.12A6.Nguyễn ...	Medium	DONE	Done	Apr 08, 2026, 10:18 PM	Apr 08, 2026, 10:18 PM
☐ KAN-9 fix: profile identity issue	Unassigned	21.12A6.Nguyễn ...	Medium	DONE	Done	Apr 08, 2026, 10:16 PM	Apr 08, 2026, 10:16 PM
☐ KAN-8 fix: add user in SciTech	Unassigned	21.12A6.Nguyễn ...	Medium	DONE	Done	Apr 08, 2026, 10:15 PM	Apr 08, 2026, 10:15 PM
☐ KAN-7 fix: faculty issue and remove admin section	Unassigned	21.12A6.Nguyễn ...	Medium	DONE	Done	Apr 08, 2026, 10:15 PM	Apr 08, 2026, 10:15 PM
☐ KAN-6 fix: incorrect score when creating new template	Unassigned	21.12A6.Nguyễn ...	Medium	DONE	Done	Apr 08, 2026, 10:02 PM	Apr 08, 2026, 10:02 PM

Picture Error: Reference source not found-19 Bug management using Jira

Bug ID	Module/Area	Bug Description (Jira Summary)	Priority	Status	Resolution
KAN-16	Homepage	Navigation issue with the "Submit now" button on the homepage (fix: home, submit now button nav).	Medium	DONE	Done
KAN-15	Navigation	Incorrect home layout navigation based on user role (fix: home layout navigation based on role).	Medium	DONE	Done
KAN-14	SciTech Admin	Missing profile page for the SciTech module (fix: scitech profile page is missing).	Medium	DONE	Done
KAN-13	Localization	Large-scale manual localization issue requiring a Python script fix (fix: big manual localization using python script).	Medium	DONE	Done
KAN-12	Localization	Missing or unsynchronized manual localization for Faculty, Author, and Co-author roles (fix: manual localization for faculty, author, co...).	Medium	DONE	Done
KAN-11	Localization	Missing or unsynchronized manual localization for the	Medium	DONE	Done

		Home and SciTech modules (fix: manual localization for home, scitech).			
KAN-10	Search/Filter	Category search functionality is not working (fix: category search not working).	Medium	DONE	Done
KAN-9	User Profile	Profile identity issue regarding user identification (fix: profile identity issue).	Medium	DONE	Done
KAN-8	SciTech Admin	Error when adding a user in the SciTech section (fix: add user in SciTech).	Medium	DONE	Done
KAN-7	Faculty Leader	Faculty issue and failure to remove redundant admin sections (fix: faculty issue and remove admin section).	Medium	DONE	Done
KAN-6	Template Config	System calculating incorrect scores when creating a new template (fix: incorrect score when creating new template).	Medium	DONE	Done

Table Error: Reference source not found-1 Defect log detail

4.2. Test Summary Metrics

Metric	Value
Total Test Cases Executed	150
Total Passed	150
Total Failed	0
Pass Rate	100%
Total Defects Logged	11

Table Error: Reference source not found-2 Test summary metrics

Based on the metrics presented in the table above, the final testing phase has concluded with highly satisfactory results, demonstrating the overall stability and reliability of the system. A comprehensive suite of 150 test cases was executed across all functional modules.

Key Takeaways from the Metrics:

Perfect Pass Rate: Out of the 150 executed test cases, all 150 successfully passed, resulting in 0 failed cases and an absolute 100% pass rate. This confirms that all tested features, navigation flows, and business rules are currently functioning exactly as specified in the system requirements.

Effective Quality Assurance: The record of 11 total defects logged represents the cumulative number of bugs identified during previous iterative testing cycles. Given that the current failure count is zero, it clearly indicates that the development team has been highly responsive, successfully resolving all previously identified issues prior to this final execution cycle.

CHAPTER 5. Conclusion & Future Work

5.1. Achievements and Compliance with Requirements

The team has successfully fulfilled 100% of the project requirements as defined in the course syllabus. A total of more than 250 non-duplicated test cases were designed, distributed evenly among four team members, and executed systematically across all major modules of the IdeaTrack system. This ensured comprehensive coverage of both functional and non-functional aspects, particularly focusing on workflow integrity and role-based access control.

From a methodological perspective, the team effectively applied ISTQB-standard testing techniques, including State Transition Testing, Equivalence Partitioning (EP), and Boundary Value Analysis (BVA). These approaches were particularly suitable for validating the complex approval workflow and input-driven functionalities within the system. In addition, the team successfully implemented a structured defect management process using JIRA Software, enabling clear tracking, classification, and prioritization of issues throughout the testing lifecycle.

Furthermore, the project achieved additional merit by integrating test automation using Selenium WebDriver with xUnit and the Page Object Model (POM). This automation effort, although limited in scope, demonstrated the team's ability to extend beyond manual testing and contributed to fulfilling the bonus point requirements of the course.

5.2. Challenges and Limitations

Despite the successful completion of the testing process, the team encountered several challenges during the project. One of the primary difficulties was the time constraint, as the system under test involved a complex multi-stage approval workflow rather than a conventional linear application such as an e-commerce platform. Understanding and validating the intricate state transitions, role dependencies, and exception flows (e.g., resubmission, revocation, and re-evaluation) required significant effort in both analysis and test design within a limited timeframe.

Another notable limitation was related to UI Automation Testing. During the implementation of Selenium-based scripts, the team faced issues with the instability of HTML/CSS selectors in the frontend layer. Frequent changes in element identifiers and page structure led to test flakiness, requiring continuous maintenance and adjustment of automation scripts. This impacted the reliability and scalability of automated test

execution, highlighting the need for more stable locator strategies or collaboration with developers for better testability support.

5.3. Suggestions for Future Enhancements

To further improve the quality assurance process and system reliability, the following recommendations are proposed for future work:

CI/CD Integration for Automated Testing: Integrate the existing automation test scripts into a CI/CD pipeline (e.g., GitHub Actions or similar tools), each code commit or pull request triggers automated test execution. This would enable continuous validation, faster feedback cycles, and early detection of regressions.

API Load and Performance Testing: Extend the testing scope to include performance and load testing using tools such as JMeter or Postman. This is particularly important for evaluating system behavior under peak conditions, such as when a large number of authors submit initiatives simultaneously near submission deadlines.

Advanced Security and Penetration Testing: Conduct comprehensive penetration testing to identify and mitigate potential vulnerabilities across the application. Given the discovery of critical issues such as Stored XSS and unrestricted file upload, a deeper security assessment is necessary to ensure the system meets production-level security standards.

Overall, the project has provided valuable practical experience in applying structured software testing methodologies to a real-world system. While the current implementation demonstrates a solid foundation, the proposed enhancements will significantly improve the robustness, scalability, and security of the IdeaTrack platform in future iterations.

REFERENCES

[1] ISTQB Foundation Level Syllabus 2018.

[2] IdeaTrack Repository: <https://github.com/trieuvn/IdeaTrack>

APPENDIX

Link to JIRA: uef-team-k5i8t9zp.atlassian.net/jira/software/projects/KAN/summary

Link to GitHub repo / Automation Script: github.com/trieuvn/IdeaTrack

Test Case Management Excel Sheet:

[https://drive.google.com/drive/folders/1npqzVpQ3dikqD8xJFTqCxecadBoEqN62?
usp=sharing](https://drive.google.com/drive/folders/1npqzVpQ3dikqD8xJFTqCxecadBoEqN62?usp=sharing)